



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

ASSIGNMENT 3

SECI1143: PROBABILITY & STATISTICAL
DATA ANALYSIS – SEMESTER 2

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1.0 INTRODUCTION

Our project, titled "Gadget Use and Its Impact on Learning Outcomes," aims to explore the effects of technology usage, particularly gadgets such as smartphones and computers, on the academic performance and learning outcomes of undergraduates. In today's society, gadgets have become indispensable tools in our daily lives, serving various purposes ranging from entertainment to educational and professional uses.

We are interested in investigating how extensively undergraduates, specifically from the Faculty of Computing at the University of Technology Malaysia (UTM), rely on gadgets in their daily routines. Our research will delve into the types of gadgets they use, the amount of time spent on these devices, and their perceptions regarding the influence of gadget use on their learning experiences. Through this study, we seek to gather data on the impact of gadget use on learning outcomes among undergraduates.

This study delves into the impact of gadget use on learning outcomes among undergraduates in the Faculty of Computing at the University of Technology Malaysia (UTM). We aim to uncover the types of gadgets primarily used for educational activities such as smartphones, tablets, and laptops the frequency of their usage for learning purposes, and the average daily hours spent on these devices. Additionally, we seek to understand students' perspectives on whether gadget use is beneficial or disadvantageous in their educational pursuits. By distributing a questionnaire via Google Form through platforms like Telegram and Whatsapp, we collected 67 responses from UTM students. Our expectation is that gadget use will be more advantageous than detrimental, providing tools for learning, access to information, and enhanced engagement in education.

2.0 DATA COLLECTION

In this project, we utilised a survey methodology to gather data, employing Google Form as our data collection tool. The survey was structured into three main sections: personal information of the respondents, their usage of gadgets and its impact on learning outcomes, and their opinions regarding gadget use and its impact on learning outcomes. Detailed questions from the Google Form survey are provided in the appendix section at the end of the project for reference and transparency in methodology.

No	Question	Answer	Level of Measurement
1	Age	• Short answer (accept integer only)	Ratio
2	Gender	• Male • Female	Nominal
3	Frequency of using gadgets for learning purpose	• Always • Often • Sometimes • Rarely • Never	Ordinal
4	Gadget primarily use for educational activities	• Smartphone • Tablet/Ipad • Laptop	Nominal
5	Operating system prefer for educational purposes	• Ios • Android • Windows • MacOS • Others	Nominal
6	Hours per day spend using gadgets for educational purposes	• 0-2 hours • 3-5 hours • 6-8 hours • 9-11 hours	Ordinal
7	How would you rate the impact of gadget use on your learning outcomes?	• 1 (Rarely) - 5 (Very often)	Ordinal
8	How much do you feel that gadget use enhances your understanding of educational material?	• 1 (Rarely) - 5 (Very often)	Ordinal

No	Question	Answer	Level of Measurement
9	How would you rate the overall effectiveness of using gadgets (e.g., smartphones, tablets, laptops) for learning purposes in comparison to traditional methods	• 1 (Rarely) - 5 (Very often)	Ordinal
10	How many times per week do you engage in collaborative learning activities facilitated by gadgets?	• 0-2 times • 3-5 times • 6-8 times • 9-11 times	Interval
11	Do you prefer to take notes using pen and paper or tablet with a stylus pen?	• Pen and paper • Tablet and stylus pen	Nominal
12	How frequently do you encounter technical issues while using gadgets for learning activities	• 1 (Rarely) - 5 (Very often)	Interval
13	Do you use YouTube for learning purposes?	• Yes • No	Nominal
14	How many educational videos do you typically watch per week on YouTube or other video-sharing platforms for learning purposes?	• Less than 5 videos • 5-10 videos • 10-20 videos • More than 20 videos	Ratio
15	Do you find it more convenient to use gadgets for learning purposes?	• Yes • No	Nominal

3.0 DATA ANALYSIS

3.1 RESPONDENT BIOGRAPHY

3.1.1 Respondents' Age

Age	Frequency
17	1
18	1
19	9
20	35
21	8
22	8
23	2
24	3
TOTAL	67

Table 3.1.1.1 Frequency Distribution of Age of Respondent

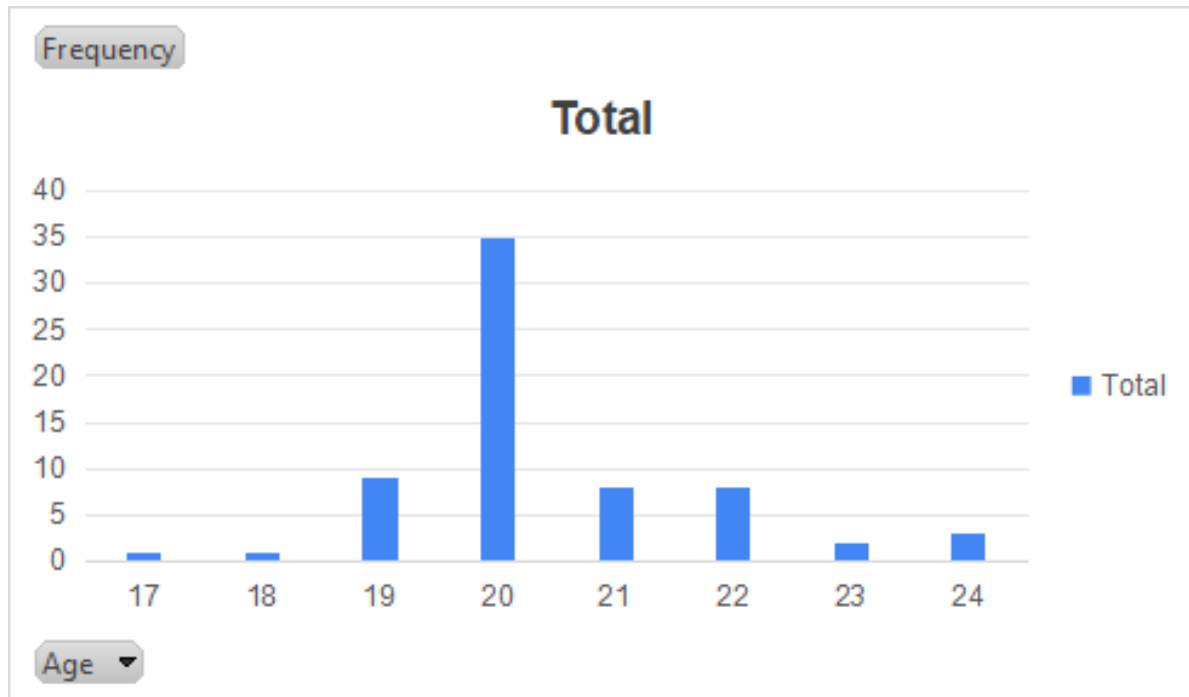


Figure 3.1.1.1 Bar Chart of Age of Respondent

Measure of Central Tendency	Value
Mean	20.4179
Mode	20
Median	20

Table 3.1.1.2 Measure of Central Tendency of Age of Respondents

As seen in Table 3.1.1.1 and Figure 3.1.1.1, the age distribution of respondents that have answered our questionnaire are around the range of age 17 to 24. Since this research is focused on undergraduates of the Faculty of Computing, UTM, most of the respondents are of age 20, with the frequency of 35. Students of age 19 come close with the frequency of 9, and there are 8 for both 21 and 22-year-old respondents. There are also 3 24-year-old and 2 23-year old respondents, and the rest, of age 17 and 18, hold a frequency of 1 student. Additionally, it can be seen from Table 3.1.1.2 that the data we gathered will mostly be from respondents of age 20, as our median and mode both are 20, and mean being 20.4179.

3.1.2 Respondent's Gender

Gender	Frequency	Percentage
Male	33	49.25%
Female	34	50.75%
TOTAL	67	100%

Table 3.1.2.1 Frequency and Percentage of Gender of Respondent

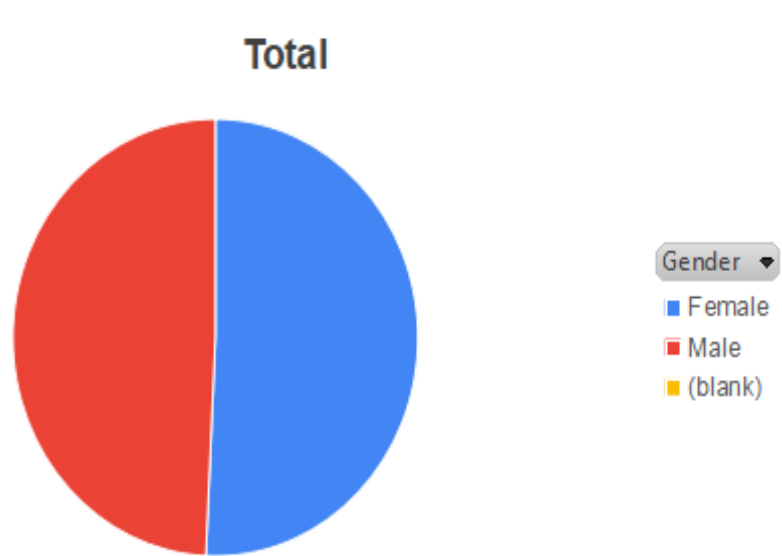


Figure 3.1.2.1 Pie Chart of Gender of Respondent

As indicated in Table 3.1.2.1 and Figure 3.1.2.1, which are the frequency table and pie chart of respondent gender, the frequency of male is 33. Meanwhile the frequency of females is 34. As a result, male has a percentage of 49.25% and female has a percentage of 50.75%. There are a total 67 respondents. In addition, the pie chart of responder gender illustrates that blue means female and red means male.

3.2 RESPONDENTS GADGET USAGE ON LEARNING PURPOSE

3.2.1 Frequency of using gadgets for learning purpose

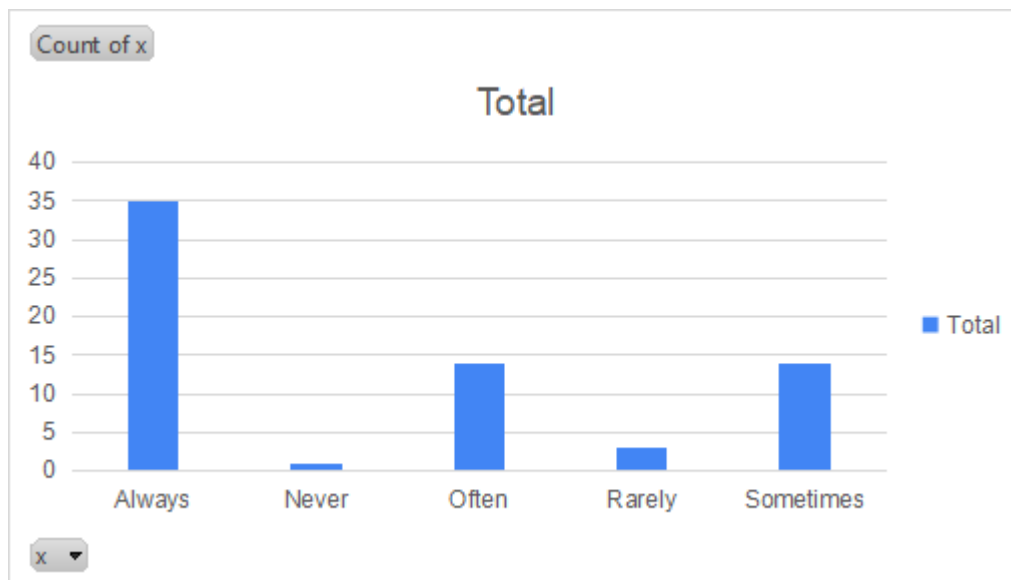


Figure 3.2.1.1 Bar Chart of Frequency of using Gadgets for Learning purpose

Adverb	Frequency
Always	35
Never	1
Often	14
Rarely	3
Sometimes	14
TOTAL	67

Table 3.2.1.1 Frequency Distribution of Frequency of using Gadgets for Learning Purpose

From Table 3.2.1.1 and Figure 3.2.1.1, we can see the frequency of students using gadgets for learning purposes varies among the surveyed population. A majority of students, 35 individuals, responded utilising gadgets "always" for educational activities, Additionally, 14 students stated that they use gadgets "often," while another 14 reported "sometimes" using them. However, there is also a smaller proportion of students who reported less frequent usage, with only 3 respondents stating they use gadgets "rarely." Then, only 1 student claimed to never use gadgets for learning purposes.

3.2.2 Gadget primarily use for Educational Activities

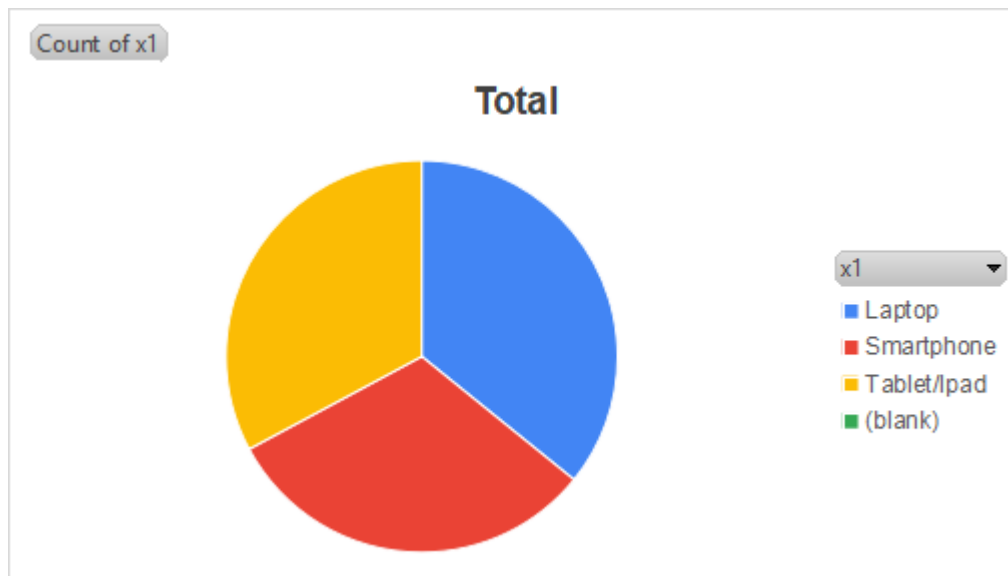


Figure 3.2.2.1 Pie Chart of Gadget primarily use by Respondents for Educational Activities

Gadgets	Frequency
Laptop	24
Smartphone	21
Tablet/Ipad	22
TOTAL	67

Table 3.2.2.1 Frequency Distribution of Gadget primarily use by Respondents for Educational Activities

In Table 3.2.2.1 and Figure 3.2.2.1, We can see the gadgets primarily being used by our respondents. The total frequency is 67. Laptop has the highest frequency of 24, which means most of the undergraduates are currently using it. Following closely behind, smartphones and tablets/iPads were also commonly utilised, with 21 and 22 students respectively reporting them as their primary educational gadgets. This data highlights that gadgets such as laptops, smartphones, and tablets/iPads being integral devices for students in their educational pursuits.

3.2.3 Operating system prefer for educational purposes

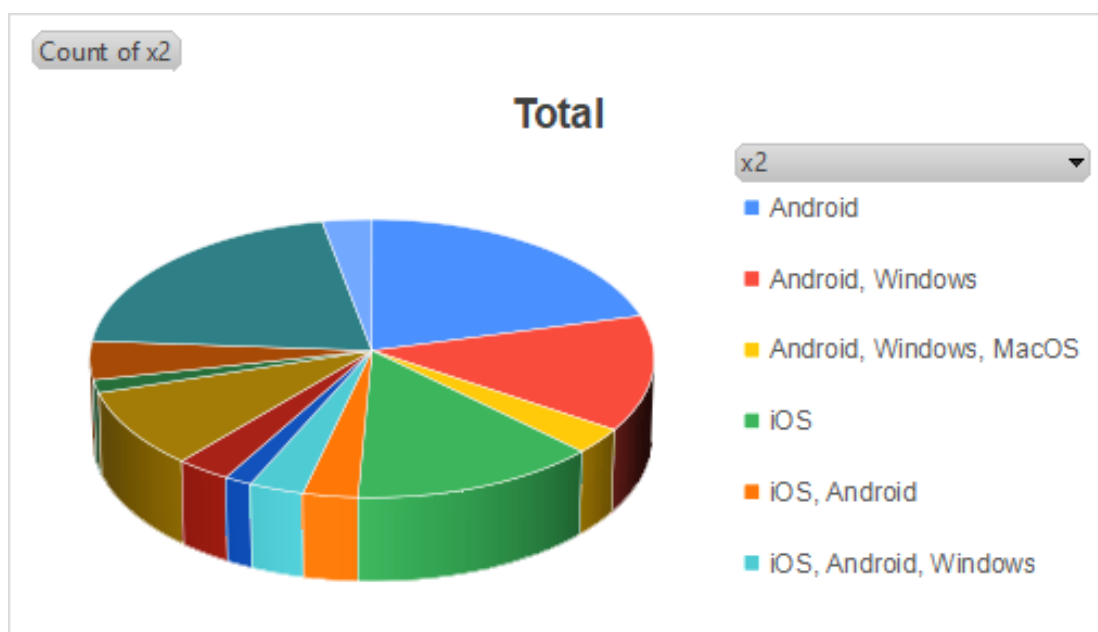


Figure 3.2.3.1 Bar Chart of Respondents OS preference for Educational Purpose

Operating System (OS)	Frequency
Android	14
Android & Windows	9
Android & Windows & MacOS	2
IOS	9
IOS & Android	2
IOS & Android & Windows	2
IOS & Android & Windows & MacOS	1
IOS & MacOS	2
IOS & Windows	6
IOS & Windows & IpadOS	1
MacOS	3
Windows	14
Windows and MacOS	2
TOTAL	67

Table 3.2.3.1 Frequency of Respondents OS preference for Educational Purpose

The data reveals a diverse preference among respondents regarding operating systems for educational purposes. Windows emerged as the most favoured choice, with 14 respondents indicating its preference. Android and iOS were also popular options, with 14 and 9 respondents respectively selecting them. Additionally, a significant number of respondents showed a preference for multiple operating systems, such as Android and Windows (9 respondents), iOS and Windows (6 respondents), and iOS and macOS (2 respondents). Interestingly, 1 respondent expressed a preference for a combination of multiple operating systems, including iOS, Android, Windows, and macOS. Overall, the data underscores the varied preferences among respondents concerning operating systems for educational purposes, reflecting the diverse technological landscape in modern education.

3.2.4 Hours per day Respondent spend using gadgets for educational purposes

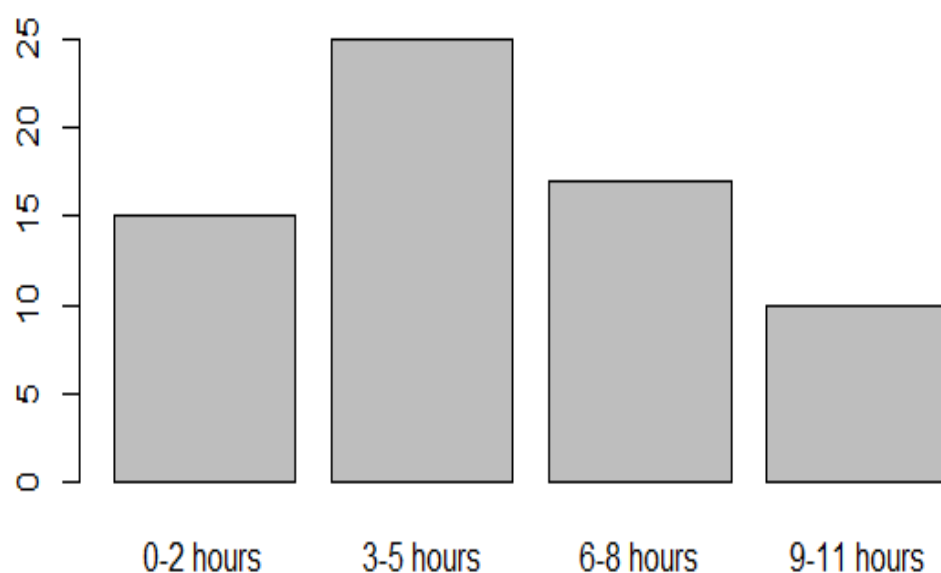


Figure 3.2.4.1 Hours per day Respondent spend using gadgets for educational purposes

Hours per day	Frequency
0-2 hours	15
3-5 hours	25
6-8 hours	17
9-11 hours	10
TOTAL	67

Table 3.2.4.1 Frequency of hours per day Respondent spend using gadgets

The data explain the distribution of respondents based on the number of hours per day they spend utilising gadgets for learning purposes. A significant portion of

respondents, comprising 25 individuals, reported spending 3-5 hours daily on educational activities with gadgets. Additionally, 17 respondents indicated spending 6-8 hours per day. Conversely, 15 respondents reported spending 0-2 hours daily. Moreover, 10 respondents reported spending 9-11 hours per day on educational activities with gadgets. Overall, the data illustrates varying levels of time allocation among respondents for utilising gadgets in their educational endeavours.

3.2.5 Impact of gadget use on Respondents' learning outcomes

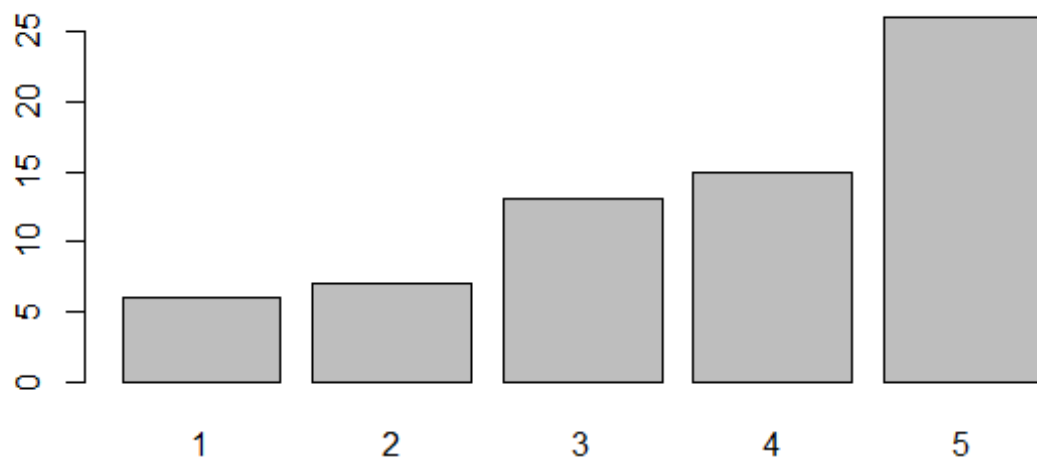


Figure 3.2.5.1 Bar Chart of Impact of Gadget use on Respondents learning outcome

Likert Scale	Frequency
Rarely	6
Seldom	7
Sometimes	13
Often	15
Very often	26
TOTAL	67

Table 3.2.5.1 Likert Scale Frequency on the Impact of Gadget use on Respondents learning outcome

The data shows how gadgets affect learning for people. Most people, 26 in total, say that gadgets "very often" help them learn better. Another 15 people say gadgets "often" help. Then, 13 people say gadgets "sometimes" help. But fewer people, 6 and 7 respectively, say gadgets "rarely" or "seldom" help them learn. Overall, it seems like many think gadgets have a positive impact on learning, especially when used a lot.

3.2.6 Respondents experience on understanding of educational material via gadgets

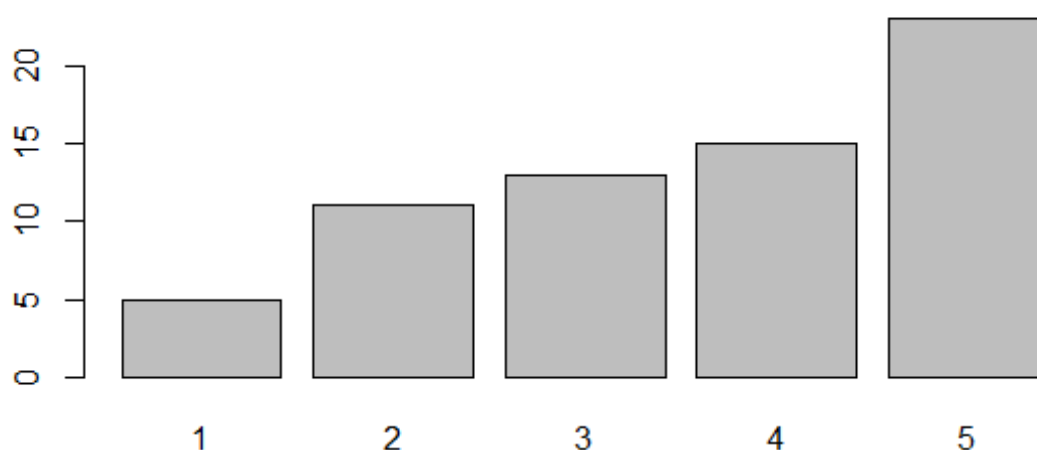


Figure 3.2.6.1 Bar Chart of Respondents experience on understanding of educational material via gadgets

Likert Scale	Frequency
Rarely	5
Seldom	11
Sometimes	13
Often	15
Very often	23
TOTAL	67

Table 3.2.6.1 Likert Scale Frequency on understanding of educational material via gadgets

The data tells us how much gadgets help people understand educational stuff. The majority, 23 people, feel like gadgets "very often" make learning easier for them. Then, 15 people say gadgets "often" help. Another 13 say gadgets "sometimes" help. But fewer people, 5 and 11 respectively, say gadgets "rarely" or "seldom" help them understand. Overall, it seems like most people believe gadgets have a positive impact on understanding educational material, especially when they're used a lot.

3.2.7 Overall Effectiveness of using gadgets compare to traditional methods

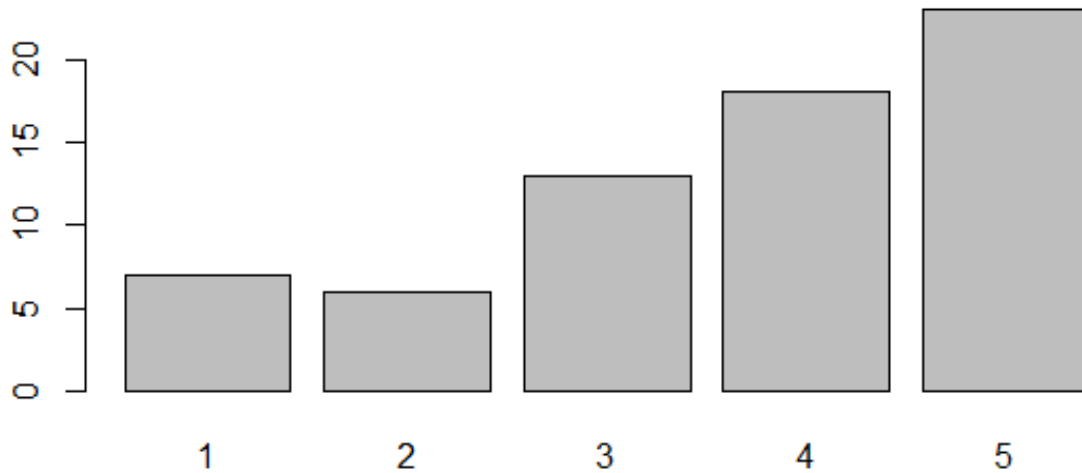


Figure 3.2.7.1 Bar Chart of overall effectiveness of using gadgets compare to traditional methods

Likert Scale	Frequency
Rarely	7
Seldom	6
Sometimes	13
Often	18
Very often	23
TOTAL	67

Table 3.2.7.1 Likert Scale Frequency on overall effectiveness of using gadgets compare to traditional methods

This data reflects how people think gadgets like smartphones, tablets, and laptops compare to traditional ways of learning. A big group, 23 people, believe gadgets are "very often" more effective than traditional methods. Then, 18 people think gadgets

"often" work better. Another 13 people say gadgets "sometimes" are more effective. But fewer people, 7 and 6 respectively, think gadgets "rarely" or "seldom" outperform traditional methods. Overall, it shows that many people find gadgets to be quite effective for learning, especially when they're used a lot.

3.2.8 Respondents engagement in collaborative learning activities facilitated by gadgets

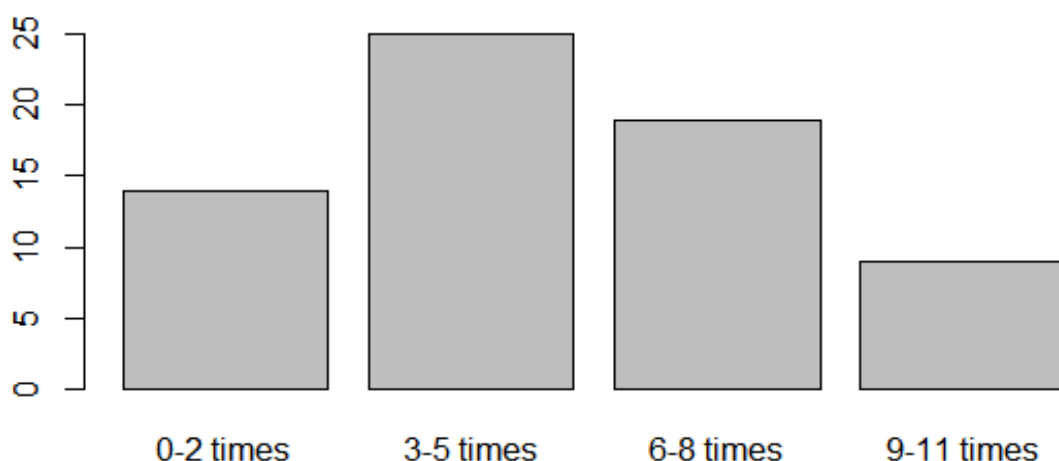


Figure 3.2.8.1 Bar Chart of respondents engagement in collaborative learning activities facilitated by gadgets

Times per week	Frequency
0-2 times	14
3-5 times	25
6-8 times	19
9-11 times	9
TOTAL	67

Table 3.2.8.1 Frequency of respondents engagement in collaborative learning activities facilitated by gadgets

This data shows how often people work together using gadgets for learning. A good number, 25 people, do collaborative learning activities using gadgets "3-5 times" a

week, showing that teamwork with technology is quite common. Following this, 19 people engage "6-8 times" a week in such activities. Additionally, 14 people do these activities "0-2 times" a week, while 9 people do them "9-11 times" a week. Overall, it suggests that collaborative learning facilitated by gadgets is so common among respondents, with the most common frequency being 3-5 times a week.

3.2.9 Do Respondents prefer to take notes using pen and paper or tablet with stylus pen?

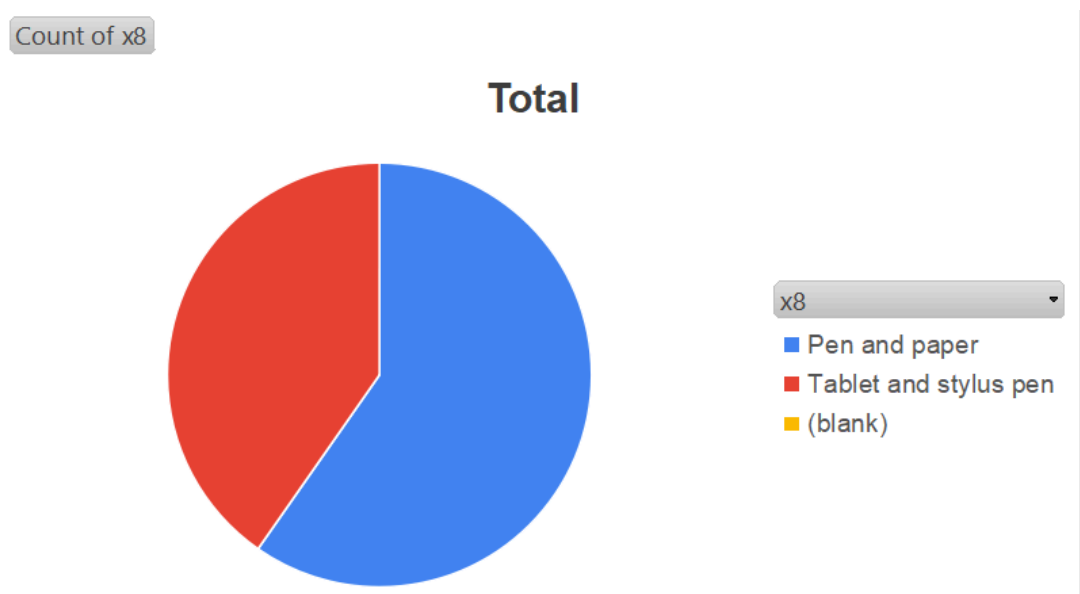


Figure 3.2.9.1 Pie Chart of Respondents answer

Notes taking Method	Frequency
Pen and Paper	40
Stylus pen and Table	27
TOTAL	67

Table 3.2.9.1 Respondents answer in Frequency

The data shows what methods people like to use for taking notes. Most people, 40 in total, prefer using pen and paper, which is the old-fashioned way. On the other hand, 27 people like using a stylus pen and tablet, which is a more modern approach. This

data tells us that while many still stick to the traditional method of writing on paper, a good number also enjoy using digital tablets for note-taking.

3.2.10 Frequency of encountering technical issues while using gadgets for learning activities

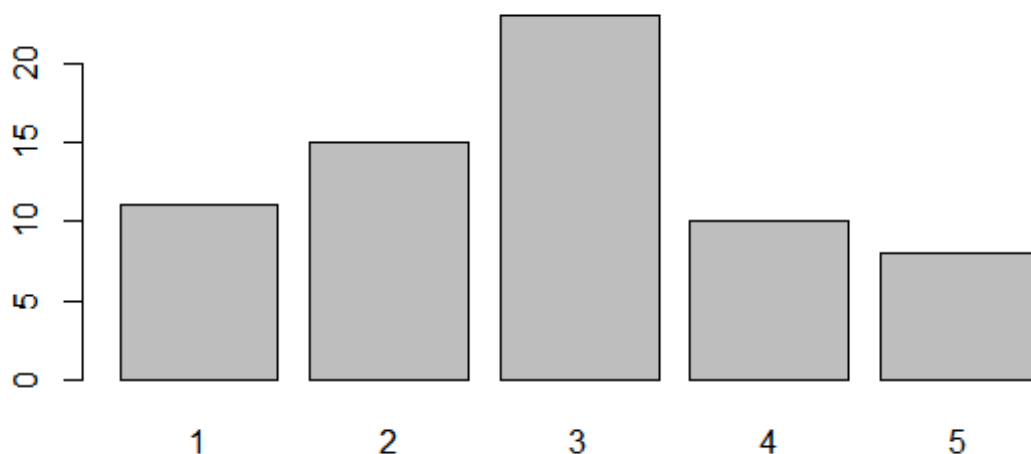


Figure 3.2.10.1 Bar Chart of Frequency of encountering technical issues while using gadgets for learning activities

Likert Scale	Frequency
Rarely	11
Seldom	15
Sometimes	23
Often	10
Very often	8
TOTAL	67

Table 3.2.10.1 Likert Scale Frequency on encountering technical issues while using gadgets for learning activities

The data provides insights into how often respondents face technical issues while using gadgets for learning activities. Among the respondents surveyed, 23 individuals reported encountering technical problems "sometimes," indicating that occasional issues like slow internet or malfunctioning devices are fairly common during their learning activities. Additionally, 15 respondents mentioned facing technical issues "seldom," suggesting that while not frequent, such problems still occur occasionally. Furthermore, 11 respondents stated that they encounter technical issues "rarely," implying that these issues are infrequent but still happen occasionally. Meanwhile, 10 respondents reported facing technical issues "often," indicating a more recurring problem. Finally, 8 respondents mentioned encountering technical issues "very often," suggesting that for a smaller group, these issues are a persistent challenge during their learning activities. Overall, the data highlights the varying degrees of frequency with which respondents experience technical issues while using gadgets for learning, with a significant portion encountering them at least occasionally.

3.2.11 Do Respondents use YouTube for learning purposes?

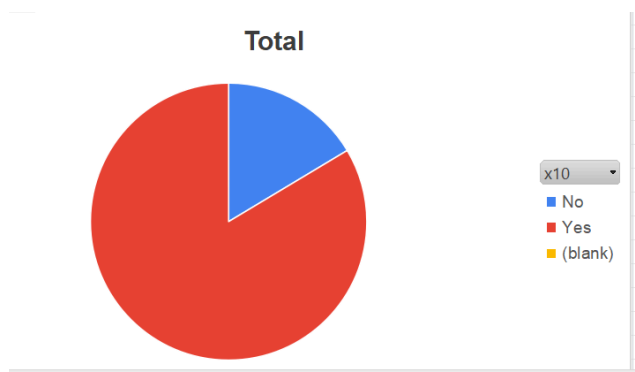


Figure 3.2.11.1 Pie Chart of respondents answer

Answer	Frequency
Yes	56
No	11
TOTAL	67

Table 3.2.11.1 Respondents answer in Frequency

The data indicates respondents' engagement with YouTube for learning purposes. A large majority of respondents, totaling 56 individuals, reported using YouTube for learning. This suggests that YouTube is a popular platform for accessing educational

content among the surveyed group. On the other hand, 11 respondents stated that they do not use YouTube for learning. While this represents a smaller portion of the respondents, it still highlights that some individuals may prefer alternative sources or methods for their learning needs. Overall, the data underscores the significant role of YouTube as a learning resource for a majority of respondents, while also recognizing that not everyone relies on this platform for educational content.

3.2.12 Number of Educational Videos Respondents watch per week on YouTube or other video-sharing platform for learning purposes

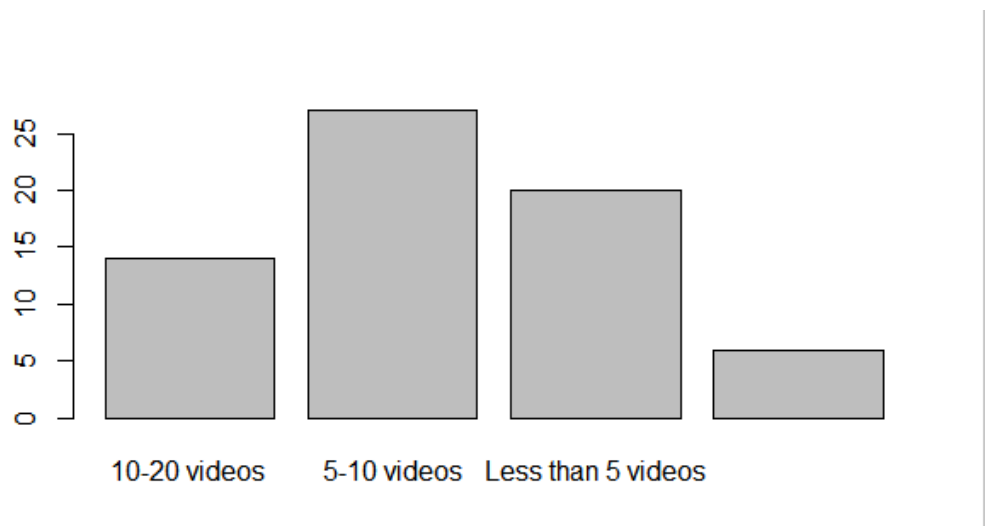


Figure 3.2.12.1 Bar Chart of Number of Educational Videos Respondents watch per week on YouTube or other video-sharing platform for learning purposes

Number of Video per week	Frequency
Less than 5 videos	20
5-10 videos	27
10-20 videos	14
More than 20 videos	6
TOTAL	67

Table 3.2.12.1 Frequency of Number of Educational Videos Respondents watch per week on YouTube or other video-sharing platform for learning purposes

The data sheds light on the frequency of video consumption among respondents for learning purposes. Among the surveyed individuals, 27 reported watching 5-10 videos per week, indicating a moderate engagement with video content for learning. Additionally, 20 respondents mentioned watching less than 5 videos per week, suggesting a smaller yet still significant portion of individuals consuming a limited number of videos for learning. Furthermore, 14 respondents stated watching 10-20 videos per week, reflecting a higher level of engagement with video-based learning materials. Conversely, 6 respondents reported watching more than 20 videos per week, indicating a smaller but notable group with a high level of video consumption for learning purposes. Overall, the data highlights the varied levels of video consumption among respondents for learning, ranging from minimal to extensive engagement with video content.

3.2.13 Do Respondents find it more convenient to use gadgets for learning purposes?

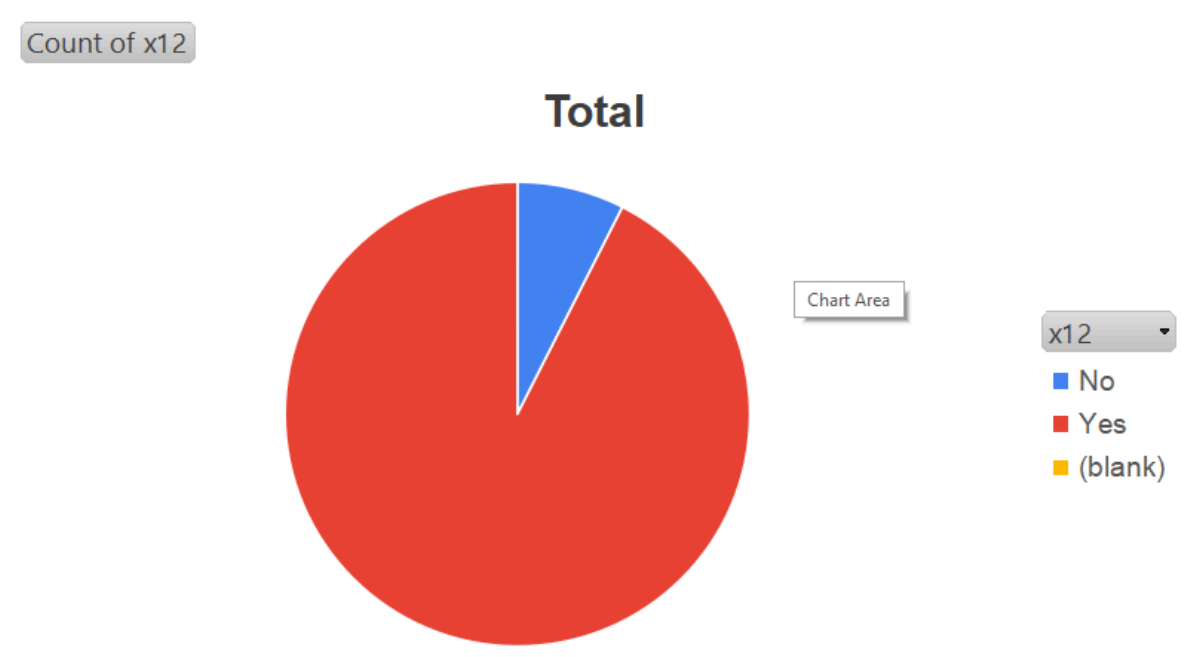


Figure 3.2.13.1 Bar Chart of Respondents answer

Answer	Frequency	Percentage
Yes	62	92.54%
No	5	7.46%
TOTAL	67	100%

Table 3.2.13.2 Respondents answer in Frequency and Percentage

The data shows that 62 respondents find it more convenient to use gadgets for learning, while only 5 respondents do not. This suggests a widespread belief among respondents that using gadgets enhances convenience in learning.

4.0 Conclusion

Our research has concluded with 67 respondents from the Faculty of Computing, University of Technology Malaysia, consisting of 33 males and 34 females. The mode age among our respondents lies at the age of 20, which is our initial target users of social media. The survey we prepared asks the respondents questions revolving around types of gadgets and its overall effectiveness. Questions such as their gadget usage time, impact on the learning outcome by gadget they use, and amount of times they use gadget in a week are among the important questions that allow us to make deductions and generalise the usage of gadgets to undergraduates' everyday.

Based on our findings, 92.54% (62 individuals) of respondents find that they feel more convenient to use gadgets for learning purposes. This overwhelming majority suggests that the majority of respondents believe that utilising gadgets enhances convenience in their learning endeavours, potentially due to the accessibility and versatility of digital learning tools. On the other hand, only 7.46% (5 individuals) of respondents stated that they do not find it more convenient to use gadgets for learning. While this represents a minority viewpoint, it still highlights that some individuals may perceive traditional methods or alternative approaches as more convenient for their learning needs. Overall, the data underscores the widespread belief among respondents that utilising gadgets for learning purposes offers enhanced convenience, indicating the prevalent adoption of digital tools in educational contexts.

In our next finding, most people think gadgets really help them learn a lot. Many people find gadgets helpful for learning because they offer easy access to a wide range of information and resources. With gadgets like smartphones, tablets, and laptops, learners can quickly search for information, access educational apps, and collaborate with others. Additionally, gadgets often provide interactive and engaging learning experiences, such as videos, simulations, and interactive quizzes, which can enhance understanding and retention of material. On the other hand, some individuals may find gadgets less helpful due to technical issues, distractions, or personal preferences for traditional learning methods. Nonetheless, the majority's belief in the effectiveness of gadgets for learning highlights their widespread use and the benefits they bring to educational pursuits.

In summary, our research journey has been a valuable learning experience, providing us with a wealth of knowledge. Initially, we learned how to create Google Forms effectively, enabling us to make clear and simple questions for our target population. We then explored ways to share our survey widely, using platforms like WhatsApp and Telegram to reach more students from the Faculty of Computing. This approach helped us get more people to participate. Additionally, we learned how to make

statistical charts using R programming, allowing us to create graphs like bar charts and pie charts with just a few lines of code. This made it easier for us to understand the data we collected. Overall, our research journey gave us useful skills and techniques that made it easier for us to conduct surveys and understand the data we collected.

We think this research experience will benefit us in the future, helping us develop both technical and communication skills. This will make us more effective employees if we're given tasks involving data collection at future companies. By getting hands-on experience with the whole process of collecting and analysing data, we'll be able to communicate more easily with our colleagues about tasks, even if they're not directly assigned to us. Lastly, we'd like to express our gratitude to our lecturer, Dr. Nor Haizan, for guiding us through this project, and to the 67 respondents who participated and helped us complete our research.

5.0 Screenshot of Google Form

Age *

Your answer

Gender *

☐ Male

☐ Female

How frequently do you use gadgets(smartphones, tablets, laptops) for learning purposes? *

☐ Always

☐ Often

☐ Sometimes

☐ Rarely

☐ Never

Which gadget do you primarily use for educational activities? *

☐ Smartphone

On a scale of 1 to 5, how would you rate the impact of gadget use on your learning outcomes? *

1 2 3 4 5

Rarely ☐ ☐ ☐ ☐ ☐ Very Often

On a scale of 1 to 5, how much do you feel that gadget use enhances your understanding of educational material? *

1 2 3 4 5

Rarely ☐ ☐ ☐ ☐ ☐ Very Often

On a scale of 1 to 5, how would you rate the overall effectiveness of using gadgets (e.g., smartphones, tablets, laptops) for learning purposes in comparison to traditional methods (e.g., textbooks, pen and paper)? *

1 2 3 4 5

Rarely ☐ ☐ ☐ ☐ ☐ Very Often

How many times per week do you engage in collaborative learning activities facilitated by gadgets (e.g., group video calls, shared documents)? *

☐ 0-2 times

☐ 3-5 times

☒ 6-8 times

Do you prefer to take notes using pen and paper or tablet with a stylus pen? *

☐ Pen and paper

☐ Tablet and stylus pen

How frequently do you encounter technical issues while using gadgets for learning activities (e.g., Wi-Fi not working, slow internet, can't login)? *

1 2 3 4 5

Rarely ☐ ☐ ☐ ☐ ☐ Very Often

Do you use YouTube for learning purposes? *

☐ Yes

☐ No

How many educational videos do you typically watch per week on YouTube or other video-sharing platforms for learning purposes? . *

☐ Less than 5 videos

☐ 5-10 videos

☐ 10-20 videos

☐ More than 20 videos

Do you find it more convenient to use gadgets for learning purposes? *

☐ Yes

☐ No

Thank you for completing our survey; your input is greatly appreciated and will contribute to our research

Your answer

5 MINUTE SURVEY, SOUNDS GOOD

SURVEY COMPLETED

Submit

Clear form

